

Payment Orchestrator

A standalone Go payment service. Single source of truth for who paid, for what, and whether access is still valid.

Stack

- **Go 1.26** — API server
 - **PostgreSQL** — payments database
 - **Razorpay** — subscription billing (INR only)
 - **chi** — HTTP router
 - **Plain HTML/CSS/JS** — UI (no build step)
-

First Time Setup

1. Install Go

Download from <https://golang.org/dl> — `go1.26.2.windows-amd64.msi`

Verify:

```
go version
# expected: go version go1.26.2 windows/amd64
```

2. Install PostgreSQL

Download from <https://www.postgresql.org/download>

Default port: 5432 Default user: `postgres`

3. Create the database and tables

```
psql -U postgres -f schema/payments.sql
```

This creates the `payments` database and all 4 tables automatically.

To verify:

```
psql -U postgres -d payments
\dt
```

You should see: `users`, `subscriptions`, `webhook_events`, `product_api_keys`

4. Configure environment

Copy `.env.example` to `.env` and fill in values:

```
cp .env.example .env
```

Minimum required to boot:

```
DB_PASSWORD=your_postgres_password
```

5. Install Go dependencies

```
go mod tidy
```

6. Run the server

```
go run main.go
```

Server starts on `http://localhost:8080`

Check health:

```
http://localhost:8080/health
```

Expected response:

```
{ "status": "ok", "mode": "skeleton" }
```

Skeleton mode is normal until Razorpay keys are added.

Environment Variables

Create a `.env` file in the project root. **Never commit this file.**

```
# Server
```

```
PORT=8080
```

```
SSO_DOMAIN=localhost:3000
```

```
# Database
```

```
DB_HOST=localhost
```

```
DB_PORT=5432
```

```
DB_USER=postgres
```

```
DB_PASSWORD=your_password_here
```

```
DB_NAME=payments
```

```
# Razorpay - fill when keys arrive
```

```
RAZORPAY_KEY_ID=
```

```
RAZORPAY_KEY_SECRET=
```

```
RAZORPAY_WEBHOOK_SECRET=
```

```
# Razorpay Plan IDs - fill after creating plans in Razorpay dashboard
```

```
PLAN_ID_PRODUCT_1_MONTHLY=
```

```

PLAN_ID_PRODUCT_1_YEARLY=
PLAN_ID_PRODUCT_2_MONTHLY=
PLAN_ID_PRODUCT_2_YEARLY=
PLAN_ID_PRODUCT_3_MONTHLY=
PLAN_ID_PRODUCT_3_YEARLY=
PLAN_ID_PRODUCT_4_MONTHLY=
PLAN_ID_PRODUCT_4_YEARLY=
PLAN_ID_PRODUCT_5_MONTHLY=
PLAN_ID_PRODUCT_5_YEARLY=
PLAN_ID_MASTER_MONTHLY=
PLAN_ID_MASTER_YEARLY=

# Per-product API keys for /verify endpoint
API_KEY_PRODUCT_1=
API_KEY_PRODUCT_2=
API_KEY_PRODUCT_3=
API_KEY_PRODUCT_4=
API_KEY_PRODUCT_5=

# CORS - add SSO origin here when integrating
ALLOWED_ORIGINS=http://localhost:8080

```

Transferring to Another Machine

Do these steps in order:

1. Copy the entire project folder
2. **Do not copy .env** — create a fresh one on the new machine
3. Install Go 1.26 on the new machine
4. Install PostgreSQL on the new machine
5. Run `psql -U postgres -f schema/payments.sql` to recreate the DB
6. Fill `.env` with the new machine's DB password
7. Run `go mod tidy` to restore dependencies
8. Run `go run main.go`

`.env` is in `.gitignore` — it will never be pushed to GitHub. Each machine needs its own `.env`.

Changing Prices or Product Names

File: `core/finance/plans.go`

- **Price:** Change `AmountPaise` — 100 paise = 1. Example: 10 = 1000

- **Product name:** Change `ProductName` only. Never change `ProductID` — it is stored in the database
- **Features:** Fill the `Features: []string{}` array per plan
- **Add a product:** Add 2 new entries (monthly + yearly). Add 2 matching env slots in `.env`

Restart the server after any change. UI reads from API — no UI changes needed.

API Endpoints

Method	Path	Purpose
GET	<code>/health</code>	Server status and mode
GET	<code>/plans</code>	All 12 SKUs
POST	<code>/subscribe</code>	Create Razorpay subscription
POST	<code>/webhook</code>	Razorpay payment confirmation (called by Razorpay directly)
GET	<code>/status/:id</code>	Poll subscription state
GET	<code>/verify</code>	Check user access (called by other products)

`/verify` usage

Called by other products to check if a user has access:

GET `/verify?email=user@example.com&product=product_1`

Header: `X-API-Key: your_product_api_key`

Response:

```
{
  "email": "user@example.com",
  "product": "product_1",
  "has_access": true,
  "role": "member",
  "reason": "active subscription"
}
```

Database Tables

Table	Purpose
<code>users</code>	Email, UUID, role (free/member/patron)
<code>subscriptions</code>	Who paid for what, Razorpay IDs, status, expiry
<code>webhook_events</code>	Idempotency log, prevents duplicate processing
<code>product_api_keys</code>	Per-product keys for <code>/verify</code>

Table	Purpose
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Useful queries:

```
-- Check all users and roles
SELECT email, role, created_at FROM users;

-- Check all subscriptions
SELECT u.email, s.plan_id, s.status, s.current_end
FROM subscriptions s
JOIN users u ON s.user_id = u.id;

-- Check webhook log
SELECT event_id, event_type, processed_at FROM webhook_events;
```

Roles

Role	Access
free	No paid access
member	Access to one specific product
patron	Access to all products (Master Plan)

Role is assigned automatically by webhook when payment is confirmed. Role is revoked automatically when subscription is cancelled or expired.

When Razorpay Keys Arrive

1. Fill `.env` — `RAZORPAY_KEY_ID`, `RAZORPAY_KEY_SECRET`, `RAZORPAY_WEBHOOK_SECRET`
2. Create 12 subscription plans in Razorpay dashboard
3. Fill the 12 `PLAN_ID_*` slots in `.env`
4. Implement the Razorpay API call in `api/subscriptions.go` where TODO Phase 6 comment sits
5. Wire Razorpay modal in `ui/checkout/index.html`
6. Restart server — zero other code changes needed

Project Structure

payment-orchestrator/

main.go	← entry point, routes, DB connection
.env	← secrets (never commit)
.env.example	← safe template to commit
go.mod / go.sum	← dependencies
.gitignore	
config/config.go	← loads all env vars
core/	
finance/	
plans.go	← all 12 SKUs - edit prices/names here
signature.go	← HMAC-SHA256 webhook verification
coupons.go	← stub, not active
entitlement/	
roles.go	← access logic, role hierarchy
api/	
helpers.go	← writeJSON, writeError
plans.go	← GET /plans
subscriptions.go	← POST /subscribe
webhook.go	← POST /webhook
status.go	← GET /status/:id
verify.go	← GET /verify
store/	
db.go	← interface contract
postgres/postgres.go	← SQL implementation
schema/payments.sql	← run this to create DB
middleware/cors.go	← CORS, named origins only
ui/	
plans/index.html	← plan selection UI
checkout/index.html	← checkout UI

What Is Parked (Not Active Yet)

- **Redis** — cache layer designed, not implemented
- **JWT** — cross-product auth designed, feature-flagged off
- **Coupons** — UI slot exists, backend stub only
- **Pro-rata upgrades** — not designed yet
- **SSO user_id** — email is current identity anchor, user_id column exists in DB ready for upgrade